

EXPERIMENT - 2

Title: Isometric Drawing using AutoCAD (2D)

1. Objective

To create isometric drawings of basic solids (cone, cylinder, and objects) using AutoCAD.

2. Software Required

AutoCAD (2D)

3. Theory (Short)

Isometric drawing is a method for visually representing 3D objects in 2D, where:

All three axes are equally inclined (120° apart)

Dimensions are drawn to scale along isometric axes

Circles appear as ellipses (isocircles)

4. Important AutoCAD Settings for Isometric Drawing

Step 1: Start AutoCAD

Open AutoCAD → New Drawing

Step 2: Set Units

Type UNITS → Enter

Set:

Type: Decimal

Insertion scale: Millimeters

Step 3: Enable Isometric Mode

Type DSETTINGS → Enter

Go to Snap and Grid tab

Under Snap Type → Select Isometric Snap

Click OK

Step 4: Switch Isoplanes

Use:

F5 or Ctrl + E

Three planes:

Isoplane Top

Isoplane Left

Isoplane Right

5. Basic Commands Used

Command	Use
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LINE	Draw edges
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ELLIPSE	Draw isocircles
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OFFSET	Parallel lines
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TRIM	Remove extra lines
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DIM	Dimensioning
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6. PROCEDURE

6.1 Drawing an Isometric Cylinder

Given Dimensions (Example)

Diameter = 40 mm

Height = 60 mm

Steps:

Switch to Isoplane Top

Type ELLIPSE → Enter

Type I (Isocircle)

Specify center → Draw circle (Ø40)

Switch to Isoplane Left/Right

Draw vertical lines (height = 60 mm)

Copy top circle upward

Join edges using LINE

Trim extra lines

6.2 Drawing an Isometric Cone

Given Dimensions

Base diameter = 40 mm

Height = 60 mm

Steps:

Set Isoplane Top

Draw base isocircle using ELLIPSE → Isocircle

Find center of base

Draw vertical axis line (height = 60 mm)

Mark apex point

Join apex to circle edges using LINE

Remove hidden lines if required

6.3 Isometric View of Object (Block Example)

Given Dimensions

Length = 80 mm

Width = 50 mm

Height = 40 mm

Steps:

Draw base rectangle using isometric lines

Use:

30° lines for width

Horizontal for length

Complete base

Draw vertical edges (height)

Join top face

Add features (holes, cuts if required)

Use isocircles for circular features

7. Dimensioning in Isometric Drawing

Step:

Type DIM

Use:

DIMLINEAR

DIMALIGNED

⚠ Note:

Dimensions are usually added in orthographic drawings, but basic dimensions can be shown for clarity.

8. Precautions

Always check correct Isoplane (F5)

Use Isocircle instead of CIRCLE

Maintain proper alignment

Avoid overlapping lines

9. Result

Isometric drawings of cone, cylinder, and object were successfully created using AutoCAD.

10. Viva Questions

What is isometric drawing?

Difference between isometric and perspective drawing

What is Isoplane?

Why is circle drawn as ellipse in isometric?

Shortcut to switch isoplane?